

## **STATEMENT OF WORK**

### **1. Objective/Requirements**

The NASA Goddard Space Flight Center (GSFC) requires the purchase of Battelle NEON Research Support Services (NRSS) to support mission-critical atmospheric and ecological research activities.

### **2. Characteristics, Scope, and Specs**

#### **Requirement is for Battelle NEON Research Support Services (NRSS).**

The vendor, Battelle Memorial Institute, shall provide the Battelle NEON Research Support Services (NRSS) with the following requirements:

#### **Scope of Work**

Battelle shall provide full operations and maintenance support for AERONET sun photometers (CIMEL CE-318N) deployed at 36 NEON-operated field sites and two NEON facilities. The purpose of this effort is to ensure uninterrupted data collection and near real-time data transfer to the AERONET data system during the Period of Performance specified herein. Services include routine preventive and corrective maintenance, minor repairs, instrument calibrations, and associated data handling in accordance with established NEON operational processes and standard procedures.

- Operations and monitoring: Support continuous operation of CIMEL CE-318N instruments at 36 NEON-operated field sites and two NEON facilities, including routine monitoring of instrument status and data flow.
- Preventive maintenance: Perform scheduled inspections, cleaning, and routine upkeep in accordance with established NEON operational processes and standard operating procedures.
- Corrective maintenance and minor repairs: Troubleshoot operational issues, replace consumables and minor components as needed, and restore instruments to nominal operation.
- Calibration support: Coordinate CIMEL calibrations with AERONET in accordance with approved workflows, including instrument handling and shipping/receiving coordination, transfer of standard calibrations, and required documentation.
- Data handling and transfer: Ensure CIMEL raw observation data are transmitted from field systems to the NEON cyberinfrastructure and subsequently transferred to AERONET in near real-time, subject to network and communications availability.
- Incident and outage response: Respond to data transfer interruptions, instrument outages, and configuration changes affecting data continuity or quality, and coordinate resolution with AERONET within mutually agreed timeframes.

- Coordination and notifications: Coordinate planned changes, outages, and configuration updates with AERONET and relevant NEON stakeholders, consistent with the Requirements section.

**Contractor-Furnished Items/Equipment:** Battelle manages a total of 49 CIMEL CE-318N instruments (36 deployed across NEON-operated field sites, one test unit at NEON HQ, three standards to support the NEON calibration facility, and 9 spare units). These instruments are Government-Owned, Contractor-Operated (GOCO) by Battelle as part of the NEON program. Battelle will not procure additional CIMEL CE-318N instruments for this project. Battelle will maintain its current inventory of consumable supplies and minor components required to operate and maintain the CIMEL instruments (e.g., spare parts, cables, mounting hardware, and cleaning supplies). NEON will use the 9 spares as rotational units for calibration and failure replacements. If any quantity of the 49 NEON-owned CIMEL instruments are deemed unserviceable, Battelle and AERONET will together decide if a replacement is required, and if necessary, AERONET will provide a replacement.

### Place of Performance

Battelle shall perform work at the NEON-operated field sites and NEON facilities listed in Table 1 as required to support operations, maintenance, calibration activities, and near real-time data transfer.

Table 1. NEON-operated sites and facilities by NEON domain (with location coordinates).

| NEON Domain | NEON Site ID | NEON Site Name                                              | Location Coordinates |
|-------------|--------------|-------------------------------------------------------------|----------------------|
| D01         | BART         | Bartlett Experimental Forest NEON                           | 44.1N, 71.3W         |
| D01         | HARV         | Harvard Forest & Quabbin Watershed NEON                     | 42.5N, 72.2W         |
| D02         | SCBI         | Smithsonian Conservation Biology Institute NEON             | 38.9N, 78.1W         |
| D02         | SERC         | Smithsonian Environmental Research Center NEON              | 38.9N, 76.6W         |
| D03         | DSNY         | Disney Wilderness Preserve NEON                             | 28.1N, 81.4W         |
| D03         | OSBS         | OSBS                                                        | 29.7N, 82.0W         |
| D04         | GUAN         | Guánica Forest NEON                                         | 18.0N, 66.9W         |
| D05         | UNDE         | University of Notre Dame Environmental Research Center NEON | 46.2N, 89.5W         |
| D06         | KONZ         | Konza Prairie Biological Station NEON                       | 39.1N, 96.6W         |
| D06         | UKFS         | University of Kansas Field Station NEON                     | 39.0N, 95.2W         |
| D07         | GRSM         | Great Smoky Mountains National Park NEON                    | 35.7N, 83.5W         |
| D07         | MLBS         | Mountain Lake Biological Station NEON                       | 37.4N, 80.5W         |
| D07         | ORNL         | Oak Ridge NEON                                              | 36.0N, 84.3W         |
| D08         | LENO         | Lenoir Landing NEON                                         | 31.9N, 88.2W         |
| D08         | TALL         | Talladega National Forest NEON                              | 33.0N, 87.4W         |

|     |        |                                                                           |               |
|-----|--------|---------------------------------------------------------------------------|---------------|
| D09 | NOGP   | Northern Great Plains Research Laboratory NEON                            | 46.8N, 100.9W |
| D09 | WOOD   | Chase Lake National Wildlife Refuge NEON                                  | 47.1N, 99.2W  |
| D10 | RMNP   | Rocky Mountains NEON                                                      | 40.3N, 105.5W |
| D10 | STER   | North Sterling NEON                                                       | 40.5N, 103.0W |
| D11 | CLBJ   | Lyndon B. Johnson National Grassland NEON                                 | 33.4N, 97.6W  |
| D11 | OAES   | Marvin Klemme Range Research Station NEON                                 | 35.4N, 99.1W  |
| D12 | YELL   | Yellowstone National Park NEON                                            | 45.0N, 110.5W |
| D13 | CPER   | Central Plains Experimental Range NEON                                    | 40.8N, 104.7W |
| D13 | CVALLA | NEON Facility - Calibration and Validation Laboratory at Longmont Airport | 40.2N, 105.2W |
| D13 | HQ     | NEON Facility – NEON Headquarters in Boulder, CO                          | 40.0N, 105.2W |
| D13 | MOAB   | Moab NEON                                                                 | 38.2N, 109.4W |
| D13 | NIWO   | Niwot Ridge NEON                                                          | 40.1N, 105.6W |
| D14 | SRER   | Santa Rita Experimental Range NEON                                        | 31.9N, 110.8W |
| D15 | ONAQ   | Onaqui NEON                                                               | 40.2N, 112.5W |
| D16 | WREF   | Wind River Experimental Forest NEON                                       | 45.8N, 122.0W |
| D17 | SJER   | San Joaquin Experimental Range NEON                                       | 37.1N, 119.7W |
| D17 | TEAK   | Lower Teakettle NEON                                                      | 37.0N, 119.0W |
| D18 | BARR   | Utqiagvik NEON                                                            | 71.3N, 156.6W |
| D18 | TOOL   | Toolik Field Station NEON                                                 | 68.7N, 149.4W |
| D19 | BONA   | Caribou-Poker Creeks Research Watershed NEON                              | 65.2N, 147.5W |
| D19 | DEJU   | Delta Junction NEON                                                       | 63.9N, 145.8W |
| D19 | HEAL   | Healy NEON                                                                | 63.9N, 149.2W |
| D20 | PUUM   | Pu'u Maka'ala Natural Area Reserve NEON                                   | 19.6N, 155.3W |

## Requirements

This section defines the minimum performance requirements for operations, communications, and data transfer associated with the AERONET sun photometers deployed at 36 NEON-operated field sites and those required as rotational spares and facility support. Battelle shall perform the requirements below in accordance with established NEON operational processes and standard operating procedures.

- Battelle and AERONET shall establish and maintain a mutually agreeable communication process for operational notifications, issue escalation, and coordination of planned changes.
- Battelle shall notify AERONET of updated or modified CIMEL instrument installations, configurations, site changes, or site decommissioning activities that could affect AERONET operations or data quality.

- Battelle and AERONET will provide calibration data, ancillary data, and instrument metadata in the agreed formats and timeframes necessary to support established data workflows.
- Raw data from CIMEL sun photometers at all NEON-operated field sites and facilities shall be transmitted from the field to the NEON cyberinfrastructure and subsequently transferred to AERONET in near real-time.
- Battelle and AERONET shall maintain synchronization between agreed-upon information stores (e.g., instrument metadata and data transfer status) on at least a daily basis subject to network or communications outages.
- In the event of a synchronization failure, designated points of contact (POCs) for AERONET and Battelle shall coordinate to restore nominal services as soon as practicable.

Field work and site activities are subject to applicable health and safety requirements and may be canceled, delayed, or subject to additional controls based on site conditions or public health guidance.

## Deliverables

Battelle shall provide the deliverables identified in Table 2 below in support of uninterrupted AERONET sun photometer operations at NEON-operated field sites and facilities. Unless otherwise specified, deliverables will be provided electronically to the AERONET Principal Investigator (or designee) and will be subject to acceptance in accordance with the Acceptance Criteria section.

Table 2. Deliverables and due dates for AERONET operations at NEON sites.

| Deliverable                          | Description                                                                                                                                                                                                                     | Frequency / Due Date                                    | Recipient |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------|
| Operational data transfer to AERONET | Near real-time transmission of CIMEL raw observation data (and associated metadata required for processing) from NEON systems to the AERONET data system, consistent with established workflows.                                | Continuous                                              | AERONET   |
| Instrument inventory and status      | Summary report of CIMEL instrument inventory and operational status, including counts across NEON operated field sites and facilities, site assignments, and any replacements from NEON spares or AERONET-provided instruments. | Annually (End of Period of Performance) or Upon Request | AERONET   |

**3. Place of Performance**



NASA Goddard Space Flight Center (GSFC)  
8800 Greenbelt Road  
Greenbelt, MD 20771

**4. Period of Performance**

May 1, 2026 – April 30, 2027